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elasticsearch_demo [2021-02-11T12:51:00,936][INFO ][o.e.n.Node           ] [8QF20ya] closing ...
elasticsearch_demo [2021-02-11T12:51:00,962][INFO ][o.e.n.Node           ] [8QF20ya] closed
elasticsearch_demo exited with code 78
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:10Z","tags":["status","plugin:kibana@6.1.0","info"],"pid":1,"state":"green","message":"Status changed from uninitialized to green - Ready","prevState":"uninitialized","prevMsg":"uninitialized"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:10Z","tags":["status","plugin:elasticsearch@6.1.0","info"],"pid":1,"state":"yellow","message":"Status changed from uninitialized to yellow - Waiting for Elasticsearch","prevState":"uninitialized","prevMsg":"uninitialized"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:10Z","tags":["status","plugin:xpack_main@6.1.0","info"],"pid":1,"state":"yellow","message":"Status changed from uninitialized to yellow - Waiting for Elasticsearch","prevState":"uninitialized","prevMsg":"uninitialized"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:10Z","tags":["status","plugin:graph@6.1.0","info"],"pid":1,"state":"yellow","message":"Status changed from uninitialized to yellow - Waiting for Elasticsearch","prevState":"uninitialized","prevMsg":"uninitialized"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:10Z","tags":["error","elasticsearch","admin"],"pid":1,"message":"Request error, retrying\nHEAD http://elasticsearch:9200/ => getaddrinfo ENOTFOUND elasticsearch elasticsearch:9200"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:10Z","tags":["status","plugin:monitoring@6.1.0","info"],"pid":1,"state":"green","message":"Status changed from uninitialized to green - Ready","prevState":"uninitialized","prevMsg":"uninitialized"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:10Z","tags":["warning","elasticsearch","admin"],"pid":1,"message":"Unable to revive connection: http://elasticsearch:9200/"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:10Z","tags":["warning","elasticsearch","admin"],"pid":1,"message":"No living connections"}
logstash_demo Sending Logstash's logs to /usr/share/logstash/logs which is now configured via log4j2.properties
logstash_demo [2021-02-11T12:51:12,617][INFO ][logstash.modules.scaffold] Initializing module {module_name=>"fb_apache",directory=>"/usr/share/logstash/modules/fb_apache/configuration"}

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Figure 8: Logstash collecting the logs and forwarding to Elasticsearch

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kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["status","plugin:reporting@6.1.0","error"],"pid":1,"state":"red","message":"Status changed from yellow to red - Unable to connect to Elasticsearch at http://elasticsearch:9200","prevState":"yellow","prevMsg":"Waiting for Elasticsearch"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["status","plugin:elasticsearch@6.1.0","error"],"pid":1,"state":"red","message":"Status changed from yellow to red - Unable to connect to Elasticsearch at http://elasticsearch:9200","prevState":"yellow","prevMsg":"Waiting for Elasticsearch"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["error","elasticsearch","data"],"pid":1,"message":"Request error, retrying\nGET http://elasticsearch:9200/_xpack => getaddrinfo ENOTFOUND elasticsearch elasticsearch:9200"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["status","plugin:security@6.1.0","error"],"pid":1,"state":"red","message":"Status changed from uninitialized to red - Unable to connect to Elasticsearch at http://elasticsearch:9200","prevState":"uninitialized","prevMsg":"uninitialized"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["security","warning"],"pid":1,"message":"Generating a random key for xpack.security.encryptionKey. To prevent sessions from being invalidated on restart, please set xpack.security.encryptionKey in kibana.yml"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["security","warning"],"pid":1,"message":"Session cookies will be transmitted over insecure connections. This is not recommended."}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["warning","elasticsearch","admin"],"pid":1,"message":"Unable to revive connection: http://elasticsearch:9200/"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["warning","elasticsearch","admin"],"pid":1,"message":"No living connections"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["warning","elasticsearch","data"],"pid":1,"message":"Unable to revive connection: http://elasticsearch:9200/"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["warning","elasticsearch","data"],"pid":1,"message":"No living connections"}
kibana_demo {"type":"log","@timestamp":"2021-02-11T12:51:13Z","tags":["license","warning","xpack"],"pid":1,"message":"License information from the X-Pack plugin could not be obtained from Elasticsearch for the [data] cluster. Error: No Living connections"}

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Figure 9: Kibana output with colour coding and status

Downloading files from a database:

You can download any file that is retrieved and displayed in the user page by clicking the 'Download' button.

Deleting files from a database: You can delete any file that is retrieved and displayed in the user page by using the 'Delete' button.

ELK logger

Gelf: The graylog extended format provides better functionality compared to the traditional syslog format. It is space optimised and reduces the payload. It has

well-defined data structures where strings and numbers can be differentiated clearly, which is absent in traditional syslog. It provides compression features vividly. We have routed the incoming traffic from Gelf to Logstash and forwarded that to Elasticsearch at port 9200.

Elasticsearch, Logstash and Kibana (ELK): We have configured the ELK stack in a Docker image with the appropriate networks and host such that it can work anywhere, provided there is an appropriate path from Gelf and Logstash.

We have written a docker-compose file for three containers. We

have to type the command *docker-compose* to start these containers. Figure 7 shows the command execution screenshot.

The three containers sync up, and the environments get configured according to the *docker-compose.yml* file. Then Logstash collects the logs, which is shown in Figure 8.

Elasticsearch and Kibana parse the input log file and colour code the incoming log message, with green being the most appropriate and correct, which is shown in Figure 9. The red coloured text is the most inappropriate and severe.